

FUNDAMENTALS OF C

C CHARACTER SET: Every programming language comes a particular character set and by using this character set only we can create the programs [software]. C comes with ASCII [American Standard Code for Information Interchange] character set, which is having 256 characters. In this **52 alphabets** [a-z, A-Z], **10 digits** [0-9], **44 operators** [+, -, *,...], **14 separators** [, . ; " " { },...] and remaining all are **special characters**.

ENGLISH LANGUAGE	C LANGUAGE
26 ALPHABETS	256 ASCII VALUES
ENGLISH WORDS	C KEYWORDS – 32
ENGLISH SENTENCES	INSTRUCTIONS
ENGLISH PARAGRAPHS	PROGRAMS
ENGLISH DOCUMENTS	SOFTWARE

Generally user ASCII values:

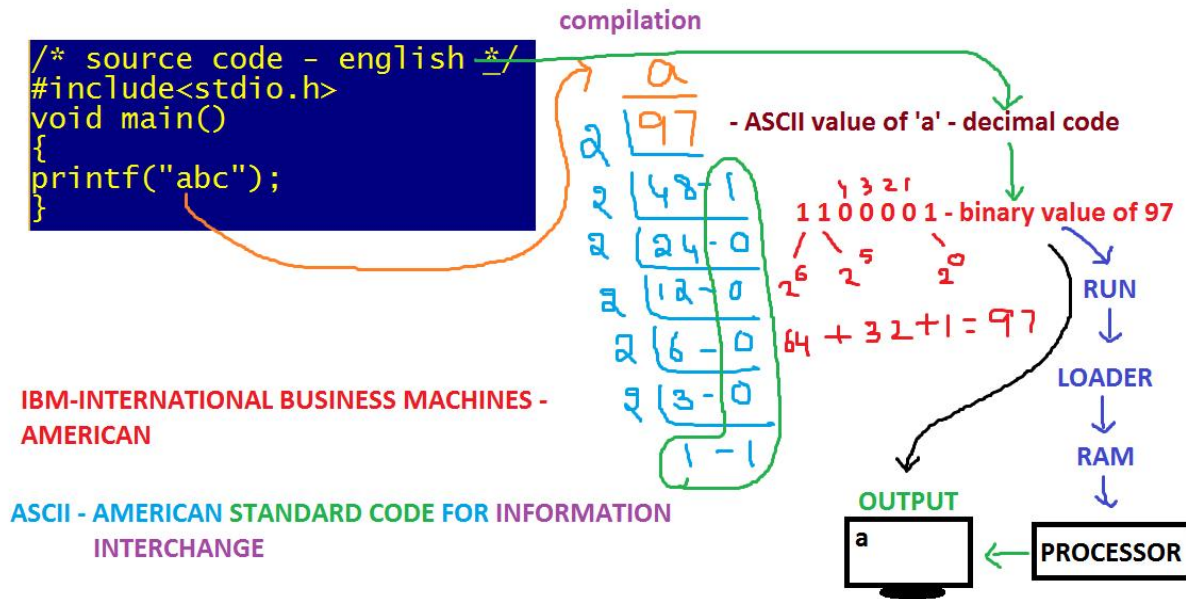
Characters	Ascii values
a-z	97-122
A-Z	65-90
0-9	48-57
*	42
+	43
Space bar	32
Tab key	9
Back space	8
Esc	27
Enter	10

C & C++ → ASCII CHARACTER SET – 256 CHAR'S

JAVA / .NET / PY – UNICODE – UNIVERSAL CODE – 65536 CHAR'S

WEB APP'S

C PROGRAM COMPILE AND EXECUTION PROCESS:



C - TOKENS

The smallest individual words we are using in developing a c program are called c – tokens. They are of different types.

- Keywords:** The system predefined / reserved words are called keywords. Each keyword is having certain meaning and we can't change this meaning. C comes with 32 keywords.
Eg: auto, break, const, continue, case, do, double, default, enum, else, for, float, goto, long, while, switch, typedef, register, static, unsigned,....
- Identifiers:** Names of variables / functions / files / array / pointers etc.

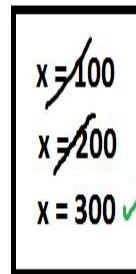
Example for variables:

int a = 10; a is used to identify the 10. Hence a is an identifier

int b = 20; b is used to identify the 20. Hence b is an identifier

Top-Down

 int x=100;
x = 200;
x = 300;
printf(x value); ==>



x is a container is used to store the values - variable

x value can be changed at any time.

change + ready

[vary + able = variable]

Example for functions:

function names



```
sum()  
{  
...;  
...;  
}
```

```
div()  
{  
...;  
...;  
}
```

Example for files:

a.c, b.cpp, c.java, d.py,...

example for array:

int a[10]; ← int array variable

example for pointer:

int * p ; int pointer variable

Identifier naming rules: